

SOA and Web Services Testing: How Different Can It Be?

August 2007

~ Underwritten, in Part, by ~

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Executive Summary

Testing and quality assurance (QA) pose unique challenges to organizations deploying service oriented architectures (SOAs) and web services applications. Unit and functional testing no longer are enough -- integration, regression, and business process testing are critical new pieces of the overall testing strategy. Add performance and security testing to the mix and you have the ingredients for significant change in the QA department.

In this study of 240 end-users, Aberdeen Group research shows that Best-in-Class (BIC) companies are taking a multi-pronged approach to the problem by incorporating automation in the testing lab and process change at the organizational level. More importantly, Best-in-Class companies are increasing the involvement of business users in all phases of the development lifecycle, and view quality not as something that's done at the end of the cycle, but as a horizontal attribute that spans multiple business processes.

Best-in-Class Performance

For this report, Aberdeen used four primary performance criteria to distinguish Best-in-Class companies from Industry Average and Laggard organizations:

- **Overall quality of deployed services:**
94% of the Best-in-Class reported an increase in software quality
- **Number of defects discovered in production code:**
61% of the Best-in-Class recorded fewer defects
- **Mean time to repair defects:**
57% of the Best-in-Class saw reduced time to repair
- **Maintainability of deployed services:**
70% of the Best-in-Class said maintainability had improved

"Traditionally we rolled out functions and features as part of a software application. Now with web services defined as standalone and atomic services, we have to be more thorough in our understanding of business requirements to translate into specific unit web services as well as to test for various scenarios to cater for most, if not all, of our users out there - both internally and externally."

~ Senior manager, retail sector

Competitive Maturity Assessment

Performance in the four KPIs used for this report is an indicator of the maturity of an organization. Typical Best-in-Class organizations:

- Track business requirements throughout the project lifecycle (81%)
- Use design-time governance as a way to establish a culture of quality (48%)
- Have deployed automated testing tools for functional tests (49%)
- Measure quality in all phases of a project, not just in QA (70%)

Required Actions

Testing assembled software in a SOA or web service environment is a complex task. Take these steps to unravel the knot:

- **Expand the focus of quality.** Take an end-to-end perspective of services and test their interoperability across an entire business process. Best-in-Class organizations track both business requirements and quality across the entire enterprise, not on a service-by-service basis.
- **Apply design-time governance.** SOA and web services promise a library of tested, documented components that can be assembled into new applications. If a developer doesn't know about a service, he or she can't use it. Move toward an institutional awareness of governance by deploying tools and policies that foster re-use.
- **Improve visibility of deployed services.** Best-in-Class organizations are using monitoring and reporting tools on their production systems to give them a better understanding of what is going wrong – and also going right – in their deployed software.

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Chapter One: Benchmarking the Best-in-Class

Aberdeen Analysis

“SOA and web service applications are un-testable,” was the comment heard recently during a discussion of new methods for testing composite applications. This new breed of software certainly poses unique challenges, especially to organizations which are slow to move away from traditional ways of testing monolithic applications. A fundamental shift in viewpoint is taking place at this time – the shift is moving away from old-style monolithic applications toward business-driven software that is assembled from components. This shift requires a parallel adjustment in not only the mechanics of software testing, but also in the approach to testing.

Maturity Class Framework

For this report, Aberdeen surveyed 240 end users involved in software quality. In analyzing the data, four primary performance criteria were used to distinguish Best-in-Class companies from Industry Average and Laggard organizations:

- The number of defects discovered in production code
- The mean time required to repair defects
- The maintainability of deployed services
- The overall quality of deployed services

With the Best-in-Class companies identified, they were then compared to the Industry Average and Laggard organizations (Table I) in a number of key areas.

Table I: Companies with Top Performance Earn Best-in-Class Status

Definition of Maturity Class	Mean Class Performance
Best-in-Class: Top 20% of aggregate performance scorers	• 61% saw a reduction in the number of defects discovered in production • 57% reported a decrease in the mean time to repair defects • 71% saw an increase in code test coverage • 94% reported an increase in the quality of deployed software

Fast Facts

- ✓ **94%** of Best-in-Class organizations report an increase in the quality of deployed services
- ✓ **61%** of Best-in-Class companies saw a reduction in the number of defects discovered in production
- ✓ **57%** of Best-in-Class reported a decrease in the time required to repair defects

Definition of Maturity Class	Mean Class Performance
Industry Average: Middle 50% of aggregate performance scorers	• 46% saw a reduction in the number of defects discovered in production • 31% reported a decrease in the mean time to repair defects • 50% saw an increase in code test coverage • 76% reported an increase in the quality of deployed software
Laggard: Bottom 30% of aggregate performance scorers	• 18% saw a reduction in the number of defects discovered in production • 21% reported a decrease in the mean time to repair defects • 13% saw an increase in code test coverage • 17% reported an increase in the quality of deployed software

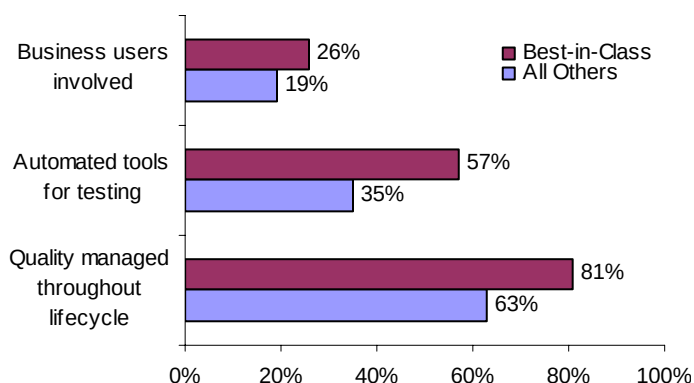
Source: Aberdeen Group, August 2007

The Best-in-Class organizations profiled in this report have attacked the problem from multiple angles. It isn't enough to just deploy automation, and it isn't enough to simply rely on functional tests – **QA for composite applications needs a horizontal, process-oriented view, not the vertical unit-test methods used in the past** (Figure 1). A director in the high-tech consulting field put it this way: "The tool-centric approach to SOA QA burned us. We had to train on the test process first before we became successful."

The typical Best-in-Class company:

- Gets business users involved in all aspects of quality
- Uses automation to increase test coverage
- Sees quality as more than just an end-of-the-lifecycle task

Figure 1: Best-in-Class Keys to Success for Quality



"QA has become a much more continuous process rather than a 'phase' in the development and deployment lifecycle."

~ VP, Software firm

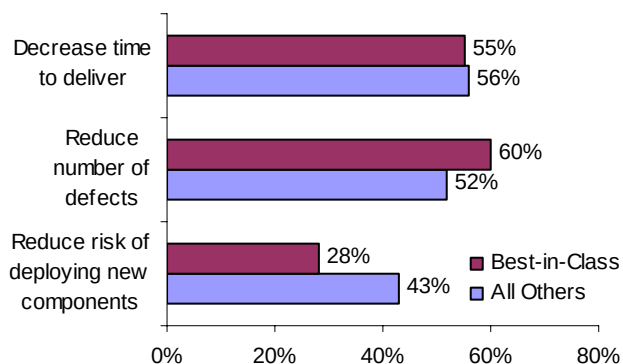
Source: Aberdeen Group, August 2007

Why Is Quality So Important?

The QA function is a large part of any software development lifecycle. Companies with formal, established QA processes often budget two to three times the length of development time for QA testing. Any inefficiency in the testing process can significantly affect the quality of the deployed product and can also have a serious impact on the overall budget.

Reducing the number of defects is important, but it isn't the top reason that most companies are looking to improve their performance in testing SOA and web services applications (Figure 2). For all groups, cost was low on the list of concerns. Only 22% of all companies surveyed indicated that reducing the cost of testing was a key driver.

Figure 2: Top Drivers for Focusing on QA

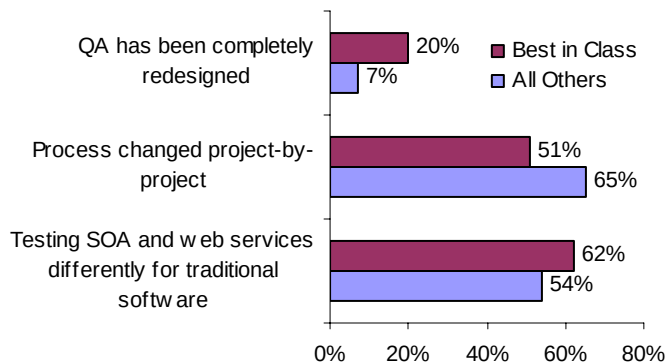


Source: Aberdeen Group, August 2007

Changing Everything

Roughly one-third of respondents to this study indicated that they had not made significant changes to their QA processes (Figure 3). Most were making incremental changes on a project-by-project basis, employing services-specific testing only when necessary.

Figure 3: Changes to Key QA Processes



Source: Aberdeen Group, August 2007

Best-in-Class companies, however, are almost **three times as likely** as Industry Average and Laggard organizations to have redesigned their entire testing process. The Best-in-Class organizations have made the commitment to services-based applications and have retooled their key processes to reflect their new direction.

Many of the changes are focused on bringing quality to bear on the entire lifecycle of the project by getting business users more involved. Doing this requires the top-down commitment of the enterprise. The enterprise as a whole must be willing to set aside the time for key players to be involved in projects as they move from phase to phase, and to supply them with the appropriate tools necessary. For example, 45% of Best-in-Class companies use tools for tracking business requirements, compared to 35% of the Industry Average and Laggard organizations.

“Traditionally we rolled out functions and features as part of a software application. Now with web services defined as standalone and atomic services we have to be more thorough in our understanding of business requirements. This allows us to translate those business requirements into specific unit web services as well as to test for various scenarios that cater for most (if not all) of our users both internally and externally,” said a senior manager in the retail sector,

The Best-in-Class PACE Model

Aberdeen uses the **PACE** model to profile the top performing 20% of companies – the Best-in-Class – in four areas:

- **P**ressures are external or internal pressures affecting an organization. These might be economic, technical, strategic, or competitive.
- **A**ctions are the ways that an organization responds to pressures.
- **C**apabilities are the internal capabilities that an organization must have in order to take the specified actions.
- **E**enablers are the technical products, procedures, or designs necessary to carry out actions.

Table 2 shows the PACE profile of the Best-in-Class companies used for this report.

Table 2: The Best-in-Class PACE Framework

Pressures	Actions	Capabilities	Enablers
Decrease the time required to deliver new services and products	- Implement automated testing tools - Involve business users in the quality process	- Quality and requirements are measured throughout the software lifecycle, not just in QA - Separate environments are in place for functional and integration testing - Key developers, business analysts, and testers are trained in new technologies	- Defect tracking and reporting applications - Production monitoring and reporting tools - Requirements tracking software - Automated orchestration and integration testing tools

Source: Aberdeen Group, August 2007

Aberdeen Insights – What is Happening to QA?

As the emphasis shifts from IT to business as the driver for technology change, the QA team is being asked to provide more than just functional testing. A functional test won't reveal how a new service interacts in an end-to-end business process. Even updating a single service carries risk and may require modeling to determine its impact. One company we spoke with does not complete incremental updates to deployed services – every change they make results in a new service in order not to upset the apple cart.

Although everyone agrees that defects and time are important, risk is a significant concern for organizations in the Industry Average and Laggard categories. For these companies that are still early on the maturity curve for SOA and web services, each additional service or component can embody a large number of dependencies and the risk of damaging running software can be high. Regression testing is done by only 39% of Laggard organizations, so the risk for them is quite real.

Best-in-Class companies tend to be further along the maturity curve and have learned the lessons of untested software. For them, the risk is known and the emphasis is instead on improving the quality of their software.

In the words of an IT manager who has been there, “Our initial experience has been tedious in defining the new processes for testing SOA and web services, with a higher learning curve for testers, however, the effort now has started paying off.”

Chapter Two: Benchmarking Requirements for Success

Best-in-Class IT organizations are changing their QA processes to encompass composite applications. These companies are also focusing on the quality of end-to-end business transactions instead of the more traditional functional testing of individual services.

Case Study: Applying Quality End-to-End at iSOFT

“QA at iSOFT has changed to implement an end-to-end view of testing and quality,” according to Phil Davies, Director of the LORENZO product unit at iSOFT. When the technology and platform of a current product had reached the limits of extendibility, the company decided to build a new product from the ground up based on SOA, instead of re-engineering the older version bit-by-bit.

The change in technical strategy was a driver for process change as well. Adopting an end-to-end view of quality meant that test plans and scripts are written and validated earlier in the lifecycle. Automation has been introduced to handle an increased volume of testing.

“It introduced other challenges. At first we learned more about the necessary process changes we needed,” said Davies. “Now we have enough instrumentation in the process to understand quantitative quality measurement, track metrics, and view improvement.”

Davies sees quality as a continuous process, where improvements are being incrementally made cycle-by-cycle. “This isn’t our end state; there is always room for improvement. For example, we need to increase the amount of tooling and automation that we use, as well as shorten the feedback time so that we can implement improvements faster.”

Fast Facts

- √ Best-in-Class companies are **twice as likely** to use automated requirements-tracking tools than Laggard organizations
- √ **70%** of Best-in-Class firms measure quality throughout the project lifecycle, not just in testing
- √ **Nearly half (47%)** of Best-in-Class companies use design-time governance to promote reuse of software components versus an average of 33% for other organizations

Competitive Assessment

The aggregated performance of surveyed companies determined whether they ranked as Best-in-Class, Industry Average, or Laggard. In addition to having common performance levels, each class also shared characteristics in five key categories: (1) process (detect and respond to changing conditions without placing additional burdens on the organization); (2) organization (corporate focus and collaboration among stakeholders); (3) knowledge management (contextualizing data and exposing it to key stakeholders); (4) technology (selection of appropriate tools and intelligent deployment of those tools); and (5) performance measurement (ability of the organization to measure the benefits of technology deployment and use the results to improve key processes further). These characteristics (Table 3) serve as a guideline for best practices and correlate directly with Best-in-Class performance across the key metrics.

Table 3: Competitive Framework

	Best-in-Class	Average	Laggards
Process	Requirements are managed throughout the software lifecycle		
	81%	72%	54%
	Automation is used for tracking business requirements		
	38%	28%	21%
Organization	Design-time governance is used to foster culture of reusability		
	48%	39%	26%
Knowledge	Key developers, business analysts, and architects are trained in new technologies		
	57%	38%	36%
Technology	Automated tools are used for functional testing SOA and web services components		
	48%	42%	29%
	Orchestration and integration testing tools are in place		
	40%	27%	19%
	Automated tools are used to extract business logic and data		
	24%	18%	13%
Performance	Quality is measured throughout the project lifecycle, not just in QA		
	70%	58%	49%

Source: Aberdeen Group, August 2007

Why QA is Coming Into Focus Now

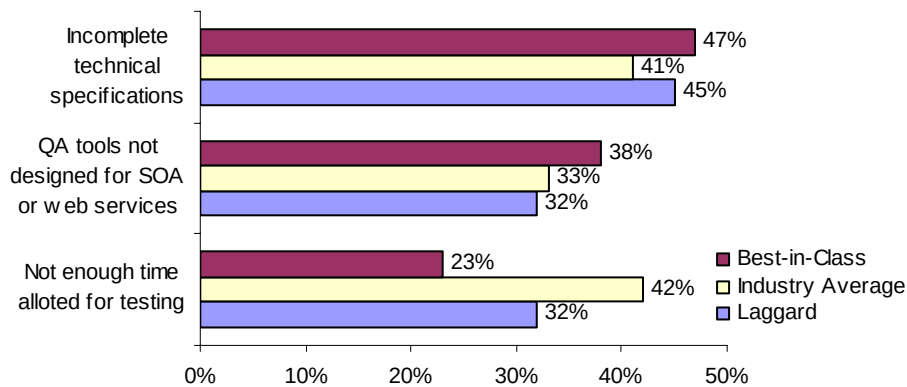
The adoption of SOA and/or web services at most companies has followed a pattern: An initial commitment is made to dedicate resources to examining SOA, followed by the establishment of a center of excellence or several pilot projects. In these scenarios, the number of deployed services is small, and the complexity of testing is low.

Research done at Aberdeen over the past two years (see Appendix B) has shown that organizations are moving now from the pilot stage to the production stage. These organizations have started deploying significant numbers of services, and are hitting a wall in terms of how they can continue to improve the quality of their operations/processes/products. What was easy in a small, self-contained pilot project is now difficult for companies in the new environment with dozens or hundreds of deployed services being used in multiple applications. It is natural to be facing hard questions about quality and risk at this stage.

The top challenge cited by Best-in-Class companies related to SOA and services testing was incomplete technical specifications, while the second

challenge was tools related. Time budgeted for QA is not an issue for the Best-in-Class, but it is the top concern of the Industry Average (Figure 4).

Figure 4: Top Challenges in QA



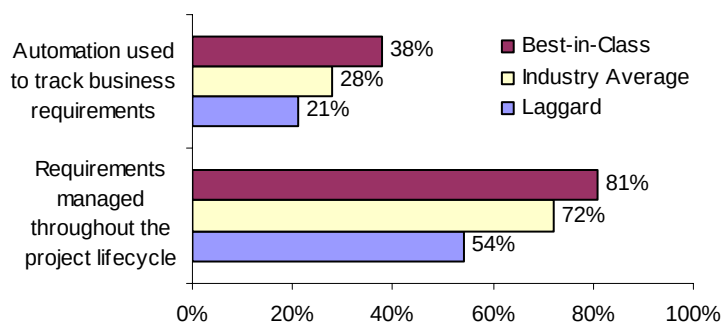
Source: Aberdeen Group, August 2007

Process

One of the key differentiators of the Best-in-Class is that they place a heavy emphasis on the business view of testing. This is consistent with Aberdeen's observation from previous reports that there has been a fundamental shift of emphasis away from IT as a principle driver of functionality, toward the business user. IT's response to this shift has been to move toward SOA and web services as a means of quickly responding to changing business requirements.

Tracking business requirements is a key component of this new way of building software. Best-in-Class companies use automation to manage those requirements across the entire lifecycle of a project, not just during the design phase (Figure 5). This capability becomes critical during regression and orchestration testing, when a component that passes unit test may not deliver the expected results from a business perspective.

Figure 5: Requirements-Based Testing Differentiates Best-in-Class Companies



Source: Aberdeen Group, August 2007

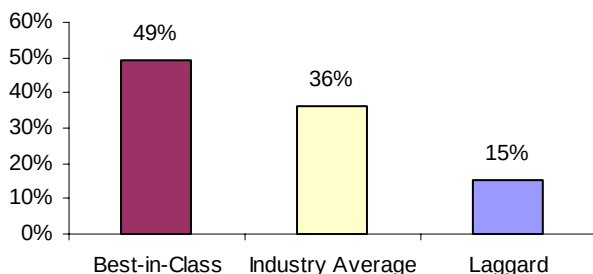
Organization

One way that Best-in-Class companies drive up the quality of deployed services is by reusing tried and tested components (Figure 6). Design-time governance is a way to build an organizational culture centered around reuse.

Although governance can be a manual process, it is more typical to have an automated toolset that matches requirements with existing code. When workflow is added, change control and approval processes are greatly simplified.

Industry Average and Laggard companies are not unaware of the advantage this brings to quality. Fifty percent of these companies state that they plan to implement design-time governance in the next twelve months.

Figure 6: Using Governance to Drive Reusability

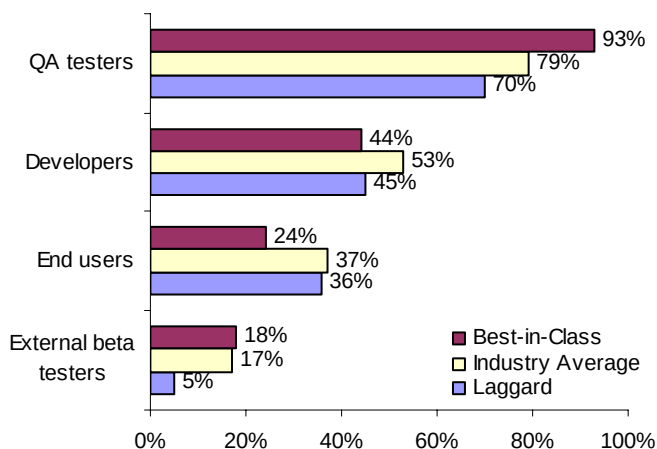


Source: Aberdeen Group, August 2007

Testing Roles

Best-in-Class companies rely more heavily on dedicated QA resources than do the other groups (Figure 7). Organizationally, these companies are further along the SOA maturity curve and are more likely to use dedicated QA teams on their projects.

Figure 7: Software Testing Roles



Source: Aberdeen Group, August 2007

Developers still play a key role in quality. For all groups it is important to emphasize that quality is not just unit and functional testing – quality in a mature organization encompasses the end-to-end assembled business process, not simply individual services.

Interestingly, when asked if organizations test SOA and web services components outside of the company, 40% of all respondents (although only 32% of Best-in-Class) said that they did. Incorporating an external service into the test plan adds complexity and risk; a trend that we will watch closely over the next several months.

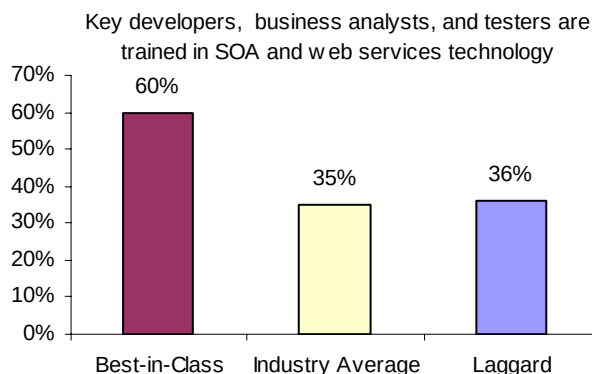
Knowledge

Training in SOA and web services technologies is an area in which Best-in-Class companies show a significant advantage (Figure 8). This isn't just about testing: Business analysts and developers must be able to think in terms of assembled applications and business processes in order to understand how to build quality into their software. Once all of the key players are aligned to a services-based architecture they can apply quality across the end-to-end business process.

"Testers are now required to know how to code as well as being part of the QA process from start to finish."

~Manager,
telecommunications
company

Figure 8: Training is an Indicator of Maturity



Source: Aberdeen Group, August 2007

Training all roles in an organization is another indicator of maturity. While Best-in-Class companies are twice as likely to have already trained their workers, both the Industry Average (57%) and Laggard (48%) companies plan to take this step in the next twelve months.

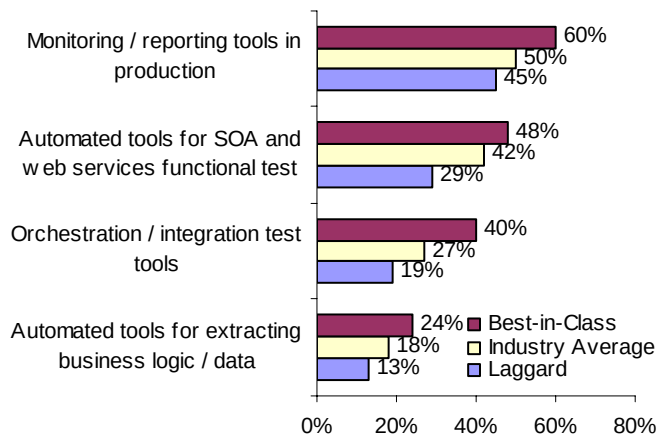
Technology

Automated testing tools are a key differentiator for Best-in-Class (Figure 9). Once the number of deployed services reaches more than a few dozen, integration and regression testing becomes difficult without the help of software.

Where and when automation become appropriate depends on the kind of testing being done. While nearly all (94%) respondents report that they perform functional testing of services (and over 80% are doing integration

testing) the numbers drop rapidly as QA becomes more complicated. While Best-in-Class companies are 50% more likely to do regression testing than Laggards (64% versus 38%), all of the groups are performing compliance (30%), orchestration (25%), and security testing (41%) at alarmingly low rates. Fifty percent of the Best-in-Class are executing performance tests.

Figure 9: Technical Approaches to SOA and Web Services QA



Source: Aberdeen Group, August 2007

"We don't have a defined strategy for testing services and the organization is not in agreement on how to test them. Some projects use free tools, some use a home-grown tool. Performance testing is not always done. Security testing really hasn't entered the picture yet either."

~Consultant, utilities industry

Aberdeen Insights – Technology

Automation is not an option when testing the complex interactions among deployed services that are part of an end-to-end transaction. Even if a single service passes unit and functional tests, the view of the quality of that service is different depending on the individual's role in the company. A service that has aced its functional tests may receive a thumbs-up from the development and QA team and still fail miserably from the business user's perspective.

Composite applications also bring new challenges to integration testing, since a single service might be used by multiple, non-connected use cases. QA and change management teams must be able to model the results of new versions of deployed services before sending them to production. While many companies have developed in-house tools to cover these areas, commercial applications are now entering the market that will assist the QA team in these increasingly complex tasks. The best will tie together functionality with requirements to provide a complete picture of the quality of a component.

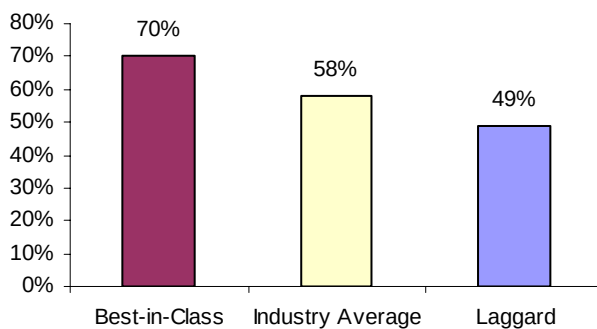
Performance

Measurement is the key to change. Best-in-Class companies are much more likely to measure the quality of a web service or SOA component across the entire development lifecycle – from the definition of business requirements to deployment and maintenance (Figure 10).

“Functionality was the main objective earlier. Now, the focus is more on how the service is exposed, invoked, its interoperability, how it is orchestrated and governed, and managing change. Assembly of services with the right version also needs to be tested. So it is now not only functionality, but a host of non-functional and technical features that needs to be tested. Testing of SOA and web services requires a different mindset,” said a vice president of IT.

In support of this, the Best-in-Class are increasing their budget for quality at a faster pace than all other respondents: 66% of the Best-in-Class indicated that they have increased their spending on QA per project versus 50% of the Industry Average and 14% of Laggards.

Figure 10: The Best-in-Class Focus on Quality Across the Entire Lifecycle



Source: Aberdeen Group, August 2007

Chapter Three: Required Actions

Based on the information in Chapters One and Two, readers should determine whether their organizations fall into the “Laggard,” “Industry Average,” or “Best-in-Class” category. The key to gaining value from this report is moving your organization along the maturity path to Best-in-Class. The following actions will help spur necessary performance improvements:

Laggard Steps to Success

- **Establish integration testing.** Take the next step in your QA journey by moving beyond unit and functional testing toward integration testing of key services. The Best-in-Class are using integration testing tools at double the rate of Laggard companies.
- **Build a culture of quality.** Best-in-Class companies treat quality as an end-to-end process rather than as the last step in the software lifecycle. Adding quality to all phases of a project will prove beneficial; start by involving the business users in both the design and the execution of test plans.
- **Automate.** The Best-in-Class are leveraging automated testing tools at double the rate of Laggards (57% versus 29%). SOA and web services are too complex to rely on manual testing alone; automation will ease the burden on the test team and allow them to focus on more strategic work, as well as significantly improve code coverage of individual components.

Industry Average Steps to Success

- **Expand the focus of quality.** Take an end-to-end perspective on services and test the services’ interoperability across the entire business process. Best-in-Class organizations track both business requirements and quality across the enterprise, not on a service-by-service basis. Use tools that provide end-to-end testing in addition to functional or interface tests.
- **Apply design-time governance.** SOA and web services provide a library of tested, documented components that can be assembled into new applications – but if a developer doesn’t know about a service, he or she can’t use it. Move toward an institutional awareness of governance by deploying tools and policies that foster re-use.
- **Improve visibility of deployed services.** Best-in-Class organizations are using monitoring and reporting tools on their production systems to give them a better understanding of what is going wrong – and what is also going right – in their applications.

Fast Facts

- ✓ **94%** of the Best-in-Class reported an increase in software quality
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- ✓ **70%** of the Best-in-Class said maintainability had improved

Best-in-Class Steps to Success

- **Increase the use of governance.** Leverage your existing design-time governance policies to increase the re-use of standardized, tested components, and then incorporating it into your change management process. Use run-time governance as an addition to your production monitoring capabilities to improve the granularity of measurement.
- **Use modeling to mitigate risk.** Modeling tools will help you uncover potential problems when upgrading or adding new services. Deploy these tools now to gain experience and insights before the need becomes critical.
- **Focus on performance.** Performance testing of composite applications is the next great frontier. As more services are pushed into production, measuring, modeling, and managing performance is becoming more critical. Don't ignore measurement of business process SLAs – in the end they will be the key indicator of a healthy application.

Aberdeen Insights – Summary

The original hypothesis for this research was that automation would be the key to successfully testing SOA and web services components. While this certainly was validated by the experiences of nearly 250 companies, a second, equally important story emerged from the data. Research showed that the best companies are testing not only vertically (using unit and functional testing as their benchmark for quality) but are also testing horizontally across an entire business process.

One key component of this approach is the involvement of the business user in more phases of the development lifecycle, and the extension of the reach of the QA team into earlier phases. While these have been characteristics of quality in other approaches to software development, they take on added power when applied to applications that are assembled from individual pieces instead of one large chunk of code.

It's the combination of people, process, and technology that make Aberdeen's Best-in-Class stand out from the crowd. These organizations understand that a tool or process taken alone isn't enough to drive quality throughout an organization.

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Appendix A: Research Methodology

Between July and August 2007, Aberdeen Group examined the responses of approximately 250 companies across a variety of geographies, industries and company revenues.

Aberdeen supplemented this online survey effort with telephone interviews with select survey respondents, gathering additional information on their QA strategies, experiences, and results.

The study aimed to identify emerging best practices for quality assurance and testing of SOA and web services-based applications, and to provide a framework by which readers could assess their own organization's capabilities and maturity.

Responding enterprises included the following:

- **Job title/function:** About two-thirds of the survey respondents are in their organizations' IT departments. The research sample included respondents with the following job titles: senior executives including CEO, COO, CFO, and VP (14%); CIO (6%); IT manager or director (37%); internal consultant (21%); and IT staff (18%).
- **Industry:** The research sample included respondents predominantly from high-technology industries and companies (representing 34% of the sample). A significant number of respondents were in the insurance industries (12%), finance and banking (18%), and telecommunications (12%).
- **Geography:** The survey respondents were distributed between North America (U.S., Canada, Mexico) 59%; Central and South America (2%); Asia/Pacific (20%); Europe, Middle East, and Africa (19%).
- **Company size:** Thirty two percent of respondents were from large enterprises (annual revenues above US\$1 billion, including 19% over US\$5 billion); 31% were from midsize enterprises (annual revenues between \$50 million and \$1 billion); and 37% of respondents were from small businesses (annual revenues of \$50 million or less).

Solution providers recognized as sponsors of this report were solicited after the fact and had no substantive influence on the direction of the report. Their sponsorship has made it possible for Aberdeen Group to make these findings available to readers at no charge.

Table 4: The PACE Framework

PACE Key
Aberdeen applies a methodology to benchmark research that evaluates the business pressures, actions, capabilities, and enablers (PACE) that indicate corporate behavior in specific business processes. These terms are defined as follows: Pressures — external forces that impact an organization's market position, competitiveness, or business operations (e.g., economic, political and regulatory, technology, changing customer preferences, competitive) Actions — the strategic approaches that an organization takes in response to industry pressures (e.g., align the corporate business model to leverage industry opportunities, such as product/service strategy, target markets, financial strategy, go-to-market, and sales strategy) Capabilities — the business process competencies required to execute corporate strategy (e.g., skilled people, brand, market positioning, viable products/services, ecosystem partners, financing) Enablers — the key functionality of technology solutions required to support the organization's enabling business practices (e.g., development platform, applications, network connectivity, user interface, training and support, partner interfaces, data cleansing, and management)

Source: Aberdeen Group, August 2007

Table 5: The Maturity Framework

Maturity Framework Key
The Aberdeen Maturity Framework defines enterprises as falling into one of the following three levels of practices and performance: Best-in-Class (20%) — Application development practices that are the best currently being employed and significantly superior to the industry norm, and result in the top industry performance. Industry norm (50%) — Application development practices that represent the average or norm, and result in average industry performance. Laggards (30%) — Application development practices that are significantly behind the average of the industry, and result in below average performance In the following categories: Process — What is the scope of process standardization? What is the efficiency and effectiveness of this process? Organization — How is your company currently organized to manage and optimize this particular process? Knowledge — What visibility do you have into key data and intelligence required to manage this process? Technology — What level of automation have you used to support this process? How is this automation integrated and aligned? Performance — What do you measure? How frequently? What's your actual performance?

Source: Aberdeen Group, August 2007

Table 6: The Relationship Between PACE and the Competitive Framework

PACE and Competitive Framework How They Interact

Aberdeen research indicates that companies that identify the most impactful pressures and take the most transformational and effective actions are most likely to achieve superior performance. The level of competitive performance that a company achieves is strongly determined by the PACE choices that they make and how well they execute.

Source: Aberdeen Group, August 2007

Appendix B: Related Aberdeen Research

Related Aberdeen research that forms a companion or reference to this report includes:

- [*The SOA in IT Benchmark Report, December, 2005*](#)
- [*The Legacy Application Migration Benchmark Report, September 2006*](#)
- [*The Composite Applications Benchmark Report, December, 2006*](#)
- [*Modernizing Legacy Application: Maximizing the Investment, June 2007*](#)
- [*SOA Middleware Starts Where Web Services Leave Off, July 2007*](#)

Information on these and any other Aberdeen publications can be found at www.Aberdeen.com.

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